Pareial DE:

I. Del- operator:

$$\nabla = \frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k} \cdots$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad \text{(laplacian)}.$$

Operator : takes in function, output result.

Linear operator: $L(c_1u_1 + c_2u_2) = c_1L(u_1) + c_2L(u_2)$.

Linear equation: $L(u) = f$ (L is Linear operator).

homo Linear : $L(u) = 0$ ($u=0$, trivial solution).

principle of superposition: if u_1, u_2 = solution of Linear homo eq, then then same for $c_1u_1 + c_2u_2$.

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2. Seperation of Var.

form applicable: $u(x,t) = \varphi(x)\,G(t)$. , $u(x,0) = f(x)$, $u(0,t) = 0$, $u(L,t) = 0$.

example 1: $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$, $u(x,0) = f(x)$, $u(0,t) = 0$, $u(L,t) = 0$.

assume $u(x,t) = \varphi(x)G(t)$.

$$\frac{\partial u}{\partial t} = \varphi(x)\cdot\frac{\partial G}{\partial t} \rightarrow \frac{\partial G}{\partial t} = k, \quad \frac{\partial \varphi}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

then $\frac{\partial u}{\partial t} = \frac{\partial}{\partial t}(\varphi(x)G(t)) = 0 + \varphi(x)\frac{\partial G(t)}{\partial t} = \varphi(x)\cdot\frac{dG}{dt} = k\frac{\partial^2 u}{\partial x^2}$

also $\frac{\partial^2}{\partial x^2}(\varphi(x)G(t)) = \frac{\partial^2\varphi}{\partial x^2}\cdot G(t) + \frac{\partial^2 G}{\partial x^2}\cdot\varphi(x) = \frac{\partial^2\varphi}{\partial x^2}\cdot G(t)$

so $\varphi(x)\frac{dG}{dt} = k\cdot\frac{\partial^2\varphi}{\partial x^2}G \rightarrow \varphi(x)\frac{dG}{dt} = k\cdot\frac{d^2\varphi}{dx^2}G$.

$$G\frac{d\varphi}{\varphi(x)} = \frac{1}{k}\frac{dG}{dt}\,dx^2$$

$$k\iint \frac{d^2\varphi}{\varphi(x)} = \iint \frac{1}{k}\frac{dG}{dt}\cdot dx^2$$

$$\int \ln(\varphi(x))\, d\varphi = \int \frac{1}{k}\frac{dG}{dt}\cdot x\, dx$$

$$\varphi\ln\varphi - \varphi = \frac{1}{2k}\frac{dG}{dt}x^2$$

$$\int u\,dv = uv - \int v\,du.$$

u	dv
$\ln\varphi$	$d\varphi$
$\frac{1}{\varphi}$	φ

$$\frac{\ln\varphi \;\big|\; d\varphi}{\varphi \;\big|\; \varphi}$$

$$\varphi\ln\varphi - \int \varphi\cdot\frac{1}{\varphi}d\varphi = \varphi\ln\varphi - \varphi$$

$$\frac{d^2\varphi}{dx^2}\,kG = \frac{dG}{dt}\cdot\varphi(x)$$

$$\frac{d^2\varphi(x)}{dx^2}\cdot\frac{1}{\varphi(x)} = \frac{dG}{dt}\cdot\frac{1}{G(t)}\cdot\frac{1}{k}$$

let $F(x) = \dfrac{d^2\varphi}{dx^2}\cdot\dfrac{1}{\varphi(x)}$, $H(t) = \dfrac{dG}{dt}\dfrac{1}{kG}$.

then $F(x) = H(t)$.

then: $\begin{cases} \dfrac{dF}{dt} = \dfrac{dF(x)}{dt} = 0 = H(t)\dfrac{dH}{dt}. \\ \dfrac{dF}{dx} = \dfrac{dF(x)}{dx} = \dfrac{dH(t)}{dx} = 0 \end{cases}$ → rate of change for both func = 0.

So $F(x) = H(t) = \lambda = \text{const.}$

let $F(x) = H(t) = -\lambda$. → $\dfrac{d^2\varphi}{dx^2} = -\lambda\varphi(x)$ ↑ eigen function!!!

$F(x)$: $\dfrac{d^2\varphi}{dx^2}\dfrac{1}{\varphi(x)} = -\lambda \rightarrow \dfrac{d^2\varphi}{dx^2} + \lambda\varphi(x) = 0$ ---①

$H(t)$: $\dfrac{1}{kG}\dfrac{dG}{dt} = -\lambda \rightarrow \dfrac{dG}{dt} + \lambda kG(t) = 0$ ----②

boundary conditions:

$u(x,0) = \varphi(x)G(0) = f(x)$, $u(L,t) = \varphi(L)G(t) = 0$. ↑ $G(t) = 0$, or $\varphi(L) = \varphi(0)$.

$u(0,t) = \varphi(0)G(t) = 0$ $\quad u(L,t) - u(0,t) = kG(t)(\varphi(L) - \varphi(0)) = 0$

since $\frac{d}{dx}\left(-\frac{d^2}{dx^2}\left(\sin\left(\frac{n\pi x}{L}\right)\right)\right)$

$= -\frac{d}{dx}\left(\cos\left(\frac{n\pi x}{L}\right)\cdot\frac{n\pi}{L}\right)$

$= \left(\frac{n\pi}{L}\right)^2 \sin\left(\frac{n\pi x}{L}\right) = \lambda \sin\left(\frac{n\pi x}{L}\right)$.

so $\sin\left(\frac{n\pi x}{L}\right)$ is an eigen ~~vecto~~ function of $\left(\frac{-d^2}{dx^2}\right)$.

since by boundary condition $\longrightarrow \left(\frac{-d^2}{dx^2}\right)$ is self-adjoint,

$\lambda_1 = \left(\frac{n\pi}{L}\right)^2$, $\lambda_2 = \left(\frac{m\pi}{L}\right)^2$. $n \neq m$.

$\lambda_1 \neq \lambda_2 \rightarrow$ ~~x≠x~~ $\longrightarrow$ eigen functions are orthogonal.

↓ for $\lambda_1 \neq \lambda_2$.

so for any two functions in $\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right)$, $\langle f_1, f_2\rangle = 0$.

let $f_1 = \sin\left(\frac{n\pi x}{L}\right)$, $f_2 = \sin\left(\frac{m\pi x}{L}\right)$. n, m are positive integers ≥ 1.

$\langle f_1, f_2\rangle = \int_0^L \sin\left(\frac{n\pi x}{L}\right)\cdot\sin\left(\frac{m\pi x}{L}\right) dx = 0$.

solution: $u(x,t) = \sum_{n=1}^{\infty} b_n e^{\left(\frac{-n\pi}{L}\right)^2 kt} \cdot \sin\left(\frac{n\pi x}{L}\right) = \sum_{n=1}^{\infty} b_n G(t)\varphi(x)$.

when $t=0 \rightarrow u(x,0) = \sum_{n=1}^{\infty} b_n \varphi_n(x) = f(x)$.

$\langle f, \varphi_m\rangle = \left\langle \sum_{n=1}^{\infty} b_n \varphi_n(x), \varphi_m \right\rangle$

$\int_0^L f\cdot \sin\left(\frac{m\pi}{L}x\right) dx = \sum_{n=1}^{\infty} b_n \int_0^L \sum_{n\neq 1}^{\infty} b_n \varphi_n(x)\cdot\varphi_m\, dx$ $= 0$ for $n \neq m$.

$\int_0^L f\cdot \sin\left(\frac{m\pi}{L}x\right) dx = \sum_{n=1}^{\infty} b_n \int_0^L \sin\left(\frac{\pi n x}{L}\right)\sin\left(\frac{m\pi x}{L}\right) dx$

$$1-2\sin^2\theta = \cos(2\theta) \rightarrow \sin^2\theta = \frac{1-\cos 2\theta}{2}$$

so for $n=m$:

$$\int_0^L f \cdot \sin\left(\frac{m\pi x}{L}\right)dx = \sum_{n=1}^{\infty} b_n \cdot \int_0^L \sin^2\left(\frac{n\pi x}{L}\right)dx$$

$$= \sum_{n=1}^{\infty} b_n \frac{1}{2}\int_0^L \left(1-\cos\left(\frac{2n\pi x}{L}\right)\right)dx$$

$$= \sum_{n=1}^{\infty} b_n \left(\frac{1}{2}L - \frac{1}{2}\left(\sin\left(\frac{2n\pi x}{L}\right)\right)\Big|_0^L\right)$$

$$\int_0^L f \cdot \sin\left(\frac{m\pi x}{L}\right)dx = \sum_{n=1}^{\infty} b_n \cdot = L \cdot \sum_{n=1}^{\infty} b_n$$

$$\int_0^L \sum_{n=1}^{\infty} b_n \int_0^L \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx = L \cdot \sum_{n=1}^{\infty} b_n.$$

$$\Downarrow$$

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right)\cdot\sin\left(\frac{m\pi x}{L}\right)dx = L$$

$$\sin\left(\frac{2n\pi \cdot L}{L}\right) - \sin\left(\frac{2n\pi \cdot 0}{L}\right) = \sin(2n\pi) - \sin(0) = 0$$

so $\sum_{n=1}^{\infty} b_n \left(\frac{1}{2}L - \left(\sin\left(\frac{2n\pi x}{L}\right)\right)\Big|_0^L\right) = \sum_{n=1}^{\infty} b_n \cdot \frac{L}{2}$

so $\int_0^L f \cdot \sin\left(\frac{m\pi x}{L}\right)dx = \sum_{n=1}^{\infty} b_n \cdot \frac{L}{2}$

$$\langle f, \varphi_m(x)\rangle = \sum_{n=1}^{\infty} b_n \cdot \frac{L}{2}$$

for $n=m \rightarrow f = b_m \varphi_m(x) \rightarrow \langle f, \varphi_m(x)\rangle = b_m \langle \varphi_m(x), \varphi_m(x)\rangle$

since $\langle \varphi_m, \varphi_m\rangle = \frac{L}{2} b_m \rightarrow \langle f, \varphi_m(x)\rangle = \frac{L}{2} b_m$.

so $b_m = \frac{2}{L}\langle \varphi_m, \varphi_m\rangle b_m = \frac{2}{L}\langle f, \varphi_m(x)\rangle$

$$= \frac{2}{L}\int_0^L f \cdot \sin\left(\frac{m\pi x}{L}\right)dx.$$

Def of self-adjoint operator:

a Linear map A, a complex vector space V.

inner product $\langle \cdot, \cdot \rangle$.

$\langle Ax, y \rangle = \langle x, Ay \rangle$ for all $x, y \in V$. if V is finite dimensional with $\perp$ basis.

prove: $\Delta = \frac{\partial^2}{\partial x^2}$ is self-adjoint operator.

since Δ map from $C \to C$ (any continuous differentiable function)

for arbitrary function f, g as function of x. on $(0, L)$

$\langle Lf, g \rangle = \int_0^L \frac{\partial^2 f}{\partial x^2} \cdot g \, dx = \int_0^L u \, dv$.

$u = \frac{\partial^2 f}{\partial x^2}$, $g dx = dv \to v = \int g dx$

prove: $L = -\frac{d^2}{dx^2}$ is self-adjoint. on $(0, L)$

$\langle Lf, g \rangle = \int_0^L \frac{-d^2 f}{dx^2} \cdot g \, dx = \int_0^L u \, dv$.

$u = g$, $dv = \frac{-d^2 f}{dx^2} \to dv = -\frac{df}{dx}$

$\int_0^L u \, dv = uv - \int v \, du$.

$\frac{du}{dx} = \frac{d(g\,dx)}{dx} = dg$

$u = \frac{-d^2 f}{dx^2}$, $dv = g \, dx \to v =$

$u = g$, $dv = \frac{-d^2 f}{dx^2} \cdot dx \to v = -\frac{df}{dx}$

$\frac{du}{dx} = \frac{dg}{dx} \to du = \frac{dg}{dx} dx$

$uv - \int v \, du = \left(-g \frac{df}{dx} \right) \Big|_0^L \int_0^L -\frac{df}{dx} \cdot \frac{dg}{dx} dx = -\left(g \frac{df}{dx} \right) \Big|_0^L + \int_0^L \frac{df}{dx} \frac{dg}{dx} dx$

$g(0)=0$, $g(L)=0$.

$$\left(-g\frac{df}{dx}\right)\Big|_0^L + \int_0^L \frac{df}{dx}\frac{dg}{dx}\,dx$$

$$= \left(-g(L)\frac{df(0)}{dx} - g(0)\frac{d(f(0))}{dx}\right) + \int_0^L \frac{df}{dx}\cdot\frac{dg}{dx}\,dx = \int_0^L \frac{df}{dx}\cdot\frac{dg}{dx}\,dx.$$

$$\langle f, L g\rangle = \int_0^L f\cdot\left(-\frac{d^2g}{dx^2}\right)dx = -\int_0^L f\cdot\frac{d^2g}{dx^2}\,dx$$

let $u = f$, $\frac{-d^2g}{dx^2}dx = dv \rightarrow v = \int \frac{-d^2g}{dx^2}\cdot dx = \frac{-dg}{dx}$

$\frac{du}{dx} = \frac{df}{dx} \rightarrow du = \frac{df}{dx}\cdot dx$

so $\left(\frac{-dg}{dx}\cdot f\right)\Big|_0^L - \int_0^L \frac{-dg}{dx}\cdot\frac{df}{dx}\,dx$

$$= \left(\frac{-dg(L)}{dx}\cdot f(L) - \frac{-dg(0)}{dx}\cdot f(0)\right) + \int_0^L \frac{dg}{dx}\frac{df}{dx}\,dx = \int_0^L \frac{dg}{dx}\frac{df}{dx}\,dx$$

so $\langle f, Lg\rangle = \langle Lf, g\rangle$.

Prove orthogonality:

Suppose φ_n, φ_m are the eigen function of operator $A = \frac{-d^2}{dx^2}$

then $A\varphi_n = \lambda_n\varphi_n$, $A\varphi_m = \lambda_m\varphi_m$

$\langle A\varphi_n, \varphi_m\rangle = \langle \varphi_n, A\varphi_m\rangle$ ← self-adjointness.

$\langle \lambda_n\varphi_n, \varphi_m\rangle = \langle \varphi_n, \lambda_m\varphi_m\rangle$

$\langle \lambda_n\varphi_n, \varphi_m\rangle - \langle \varphi_n, \lambda_m\varphi_m\rangle = 0$

$\lambda_n\langle \varphi_n, \varphi_m\rangle - \lambda_m\langle \varphi_n, \varphi_m\rangle = 0$

$(\lambda_n - \lambda_m)\langle \varphi_n, \varphi_m\rangle = 0$.

→ so for any $\lambda_n \neq \lambda$, $\langle \varphi_n, \varphi_m\rangle = 0$.

so eigenfunction is orthogonal.

$G(t) = 0$ is trivial.

so if $G(t) \neq 0$, then $\varphi(0) = 0$, $\varphi(L) = \varphi(0) = 0$.

then ① : $\frac{d^2\varphi}{dx^2} + \lambda\varphi(x) = 0$, $\varphi(0) = \varphi(L) = 0$.

$$\Downarrow$$

$$\varphi'' + \lambda\varphi = 0 \rightarrow m^2 + \lambda = 0 \rightarrow m = \pm\sqrt{-\lambda} = \pm\sqrt{\lambda}i$$

then solution $= C_0 e^{\sqrt{-\lambda}x} + C_1 e^{-\sqrt{-\lambda}x}$

$\varphi(0) = C_0 + C_1 = 0$ solution : $\varphi = C_0\cos(\sqrt{\lambda}x) + C_1\sin(\sqrt{\lambda}x) + C_2\cos(-\sqrt{\lambda}x) + C_3\sin(-\sqrt{\lambda}x)$

$\varphi(L) = C_0 e^{\sqrt{\lambda}L} = 0$

$$\varphi = C_1\cos(\sqrt{\lambda}x) + C_2\sin(\sqrt{\lambda}x)$$

$\varphi(0) = C_1 = 0 \rightarrow \varphi = C_2\sin(\sqrt{\lambda}x)$

$\varphi(L) = C_1\cos(\sqrt{\lambda}L) + C_2\sin(\sqrt{\lambda}L) = 0 \rightarrow \sqrt{\lambda}L = \pi + k\cdot\pi = n\pi$

$\sqrt{\lambda} = \frac{n\pi}{L}$

if $\lambda = 0$, $\varphi = C_0 + 0 = 0$. trivial.

if $\lambda < 0$, $-\lambda > 0 \rightarrow \pm\sqrt{-\lambda} = \text{real}$.

then solution : $\varphi = C_0 e^{-\sqrt{-\lambda}x} + C_1 e^{+\sqrt{-\lambda}x}$

$\varphi(0) = C_0 + C_1 = 0 \rightarrow C_1 = -C_0$

$\varphi(L) = C_0 e^{-\sqrt{-\lambda}L} + C_1 e^{+\sqrt{-\lambda}L} = 0$

$= C_0 e^{-\sqrt{-\lambda}L} - C_0 e^{\sqrt{-\lambda}L} = C_0\left(e^{-\sqrt{-\lambda}L} - e^{\sqrt{-\lambda}L}\right) = 0$

so $C_0 = 0 = C_1$

so trivial, so $\lambda < 0$.

$$\varphi(x) = A\sin\left(\frac{n\pi}{L}x\right)$$

$$\frac{dG}{dt}\cdot\frac{1}{kG} = -\lambda$$

$$\frac{dG}{dt} + \lambda kG = 0.$$

$$\lambda = 0 \rightarrow \frac{dG}{dt} = 0 \rightarrow G(t) = \text{const} =$$

$$m + k\lambda = 0 \rightarrow m = -\lambda k \quad G = Ce^{-\lambda kt}$$

$$\lambda k$$

$$\text{So } u(x,t) = \varphi(x)\,G(t) = C_2 \sin\left(\frac{n\pi x}{L}\right)\cdot C_0 e^{-\lambda k t}.$$

$$u(x,0) = C_2 C_0 \sin\left(\frac{n\pi x}{L}\right) = f(x)$$

$$\lambda = \left(\frac{n\pi}{L}\right)^2 \rightarrow \quad G(t) = C_0 e^{-\left(\frac{n\pi}{L}\right)^2 kt}$$

$$\text{So } u(x,t) = C_0 C_2 e^{-\left(\frac{n\pi}{L}\right)^2 kt}\cdot \sin\left(\frac{n\pi x}{L}\right)$$

$$\boxed{u(x,t) = \sum_{n=1}^{\infty} b_n\, e^{-\left(\frac{n\pi}{L}\right)^2 kt}\cdot \sin\left(\frac{n\pi x}{L}\right).}$$

(infinite many integers n that satisfy the boundary condition).

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function space → a sub-category of vector space.

close under (add - scalar x.), contains 0.

$$u(x,t) = \sum_{n=1}^{\infty} b_n \cdot \sin\left(\frac{n\pi x}{L}\right)\cdot e^{-\left(\frac{n\pi}{L}\right)^2 kt}$$

$$b_n = \frac{2}{L}\int_0^L f\cdot \sin\left(\frac{n\pi x}{L}\right) dx.$$

coordinate projection.

$$\left(f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)\right).$$

for $\begin{cases} u(x,0) = f(x). \\ u(0,t) = 0. \\ u(L,t) = 0. \end{cases}$

calculate b_n

given initial condition $f(x)$